

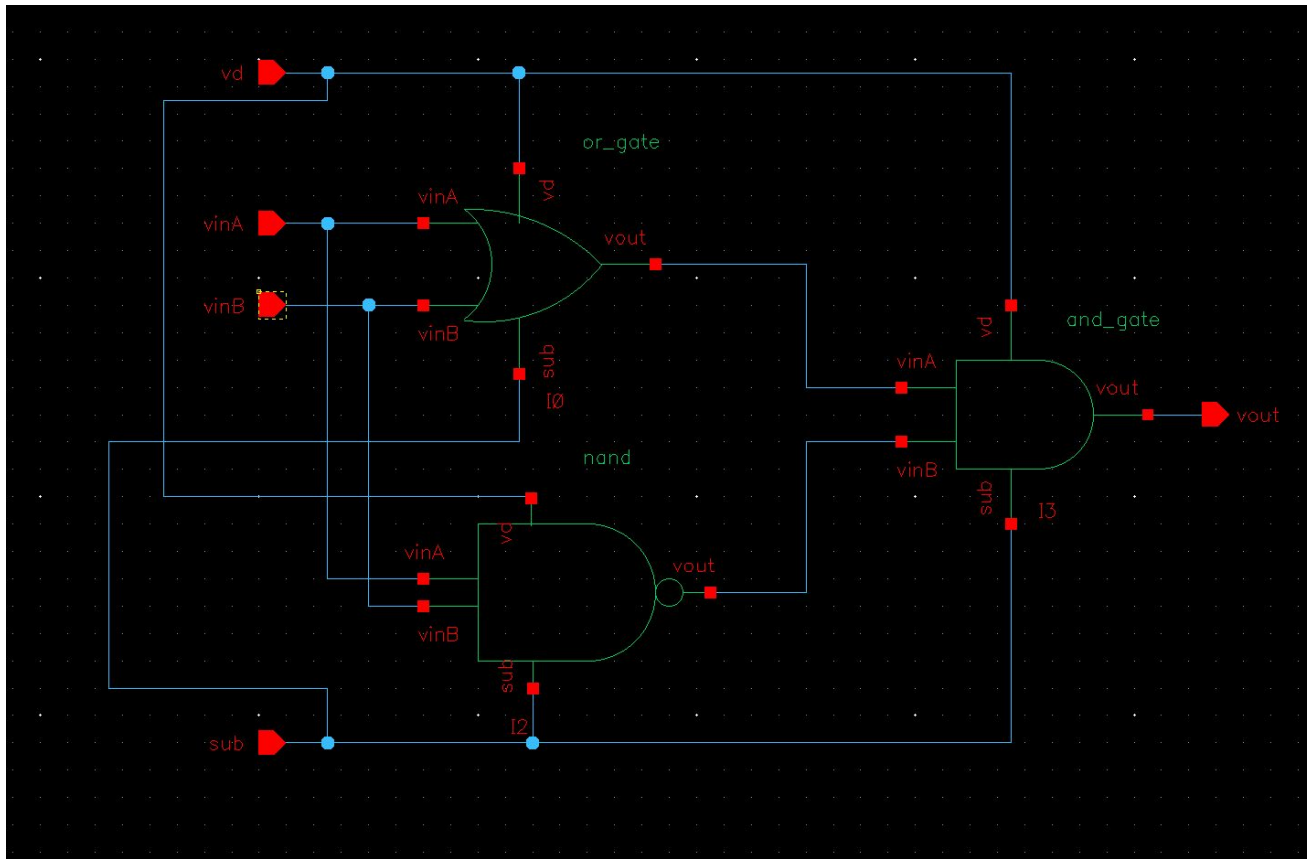


XOR-Encryptor/ Decryptor Block Layout

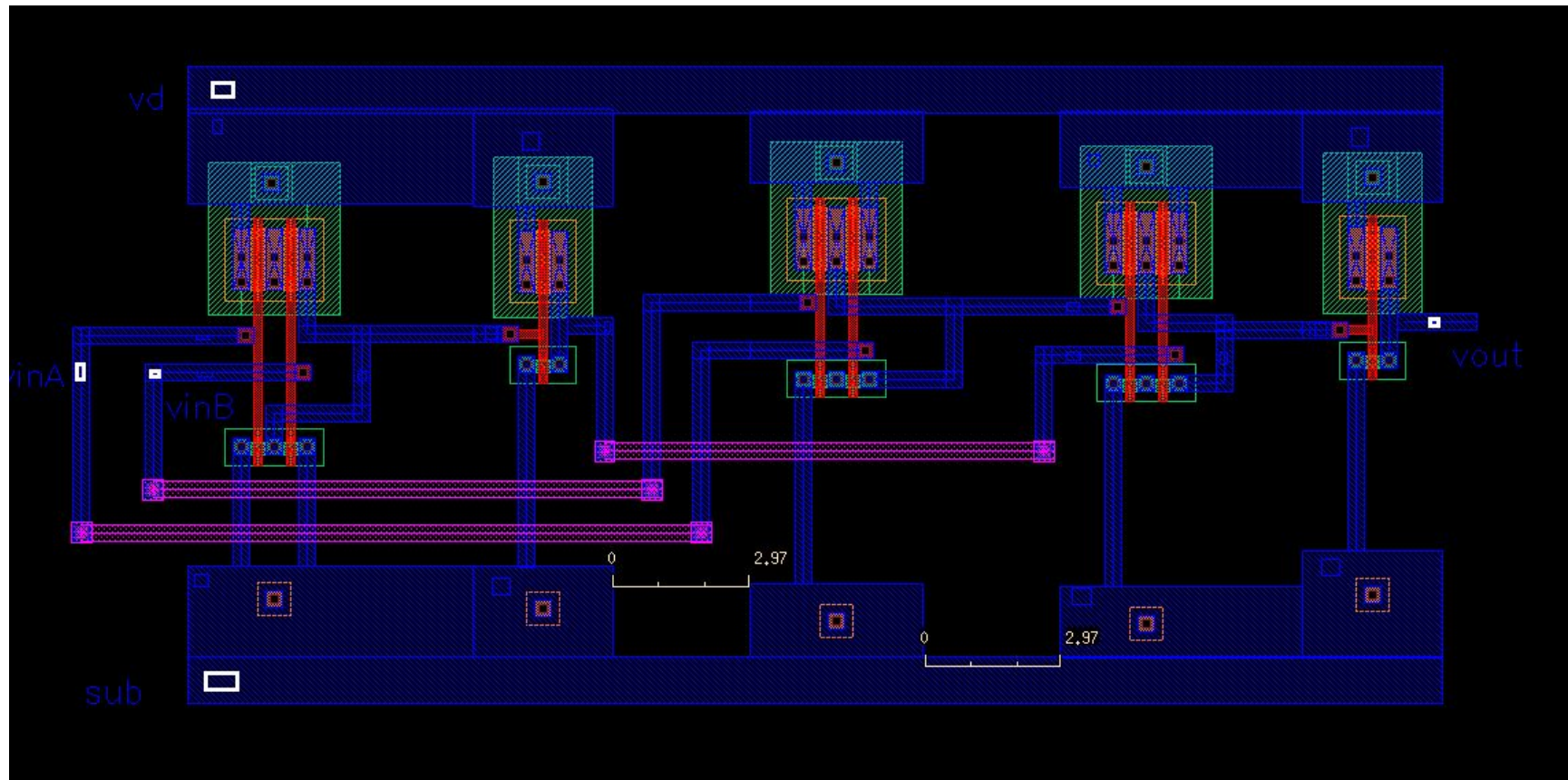
Gerald Stanford



Xor Gate



Xor gate Schematic



Xor gate Layout

```
DRC started at Wed Dec 6 21:00:49 2023
```

```
Validating hierarchy instantiation for:
```

```
library: Stanford
```

```
cell: xor_gate
```

```
view: layout
```

```
Rules come from library NCSU_TechLib_tsmc02d.
```

```
Rules path is divaDRC.rul.
```

```
Inclusion limit is set to 1000.
```

```
Running layout DRC analysis
```

```
Flat mode
```

```
Full checking.
```

```
DRC started.....Wed Dec 6 21:00:49 2023
```

```
completed ....Wed Dec 6 21:00:49 2023
```

```
CPU TIME = 00:00:00 TOTAL TIME = 00:00:00
```

```
***** Summary of rule violations for cell "xor_gate layout" *****
```

```
Total errors found: 0
```

Xor gate DRC

```

Extraction started at Wed Dec  6 21:03:51 2023

Validating hierarchy instantiation for:
library: Stanford
cell:    xor_gate
view:    layout
Rules come from library NCSU_TechLib_tsmc02d.
Rules path is divaEXT.rul.
Inclusion limit is set to 1000.
Running layout Extraction analysis
flat mode
Full checking.
For layer nChannelTran :
  8 shapes encountered.
  8 nmos4 ivpcell NCSU_Analog_Parts devices well formed.
For layer pChannelTran :
  8 shapes encountered.
  8 pmos4 ivpcell NCSU_Analog_Parts devices well formed.
0 cap ivpcell NCSU_Analog_Parts parasitics created.
0 cap ivpcell NCSU_Analog_Parts parasitics created.
0 cap ivpcell NCSU_Analog_Parts parasitics created.
0 cap ivpcell NCSU_Analog_Parts parasitics created.
0 cap ivpcell NCSU_Analog_Parts parasitics created.
0 cap ivpcell NCSU_Analog_Parts parasitics created.
Extraction started.....Wed Dec  6 21:03:51 2023
      completed ....Wed Dec  6 21:03:52 2023
      CPU TIME = 00:00:00  TOTAL TIME = 00:00:01
***** Summary of rule violations for cell "xor_gate layout" *****
      Total errors found: 0

saving rep Stanford/xor_gate/extracted
Getting layout property bagGetting layout property bag
*Error* eval: unbound variable - ivLVSContentsDifferForm
*WARNING* ddGetObj: cell name 'xor_gate' is not a legal cell name.
*WARNING* ddGetObj: cell name 'xor_gate' is not a legal cell name.
*WARNING* ddGetObj: cell name 'xor_gate' is not a legal cell name.
*WARNING* ddGetObj: cell name 'xor_gate' is not a legal cell name.
LVS job is now started...
The LVS job has completed. The net-lists match.

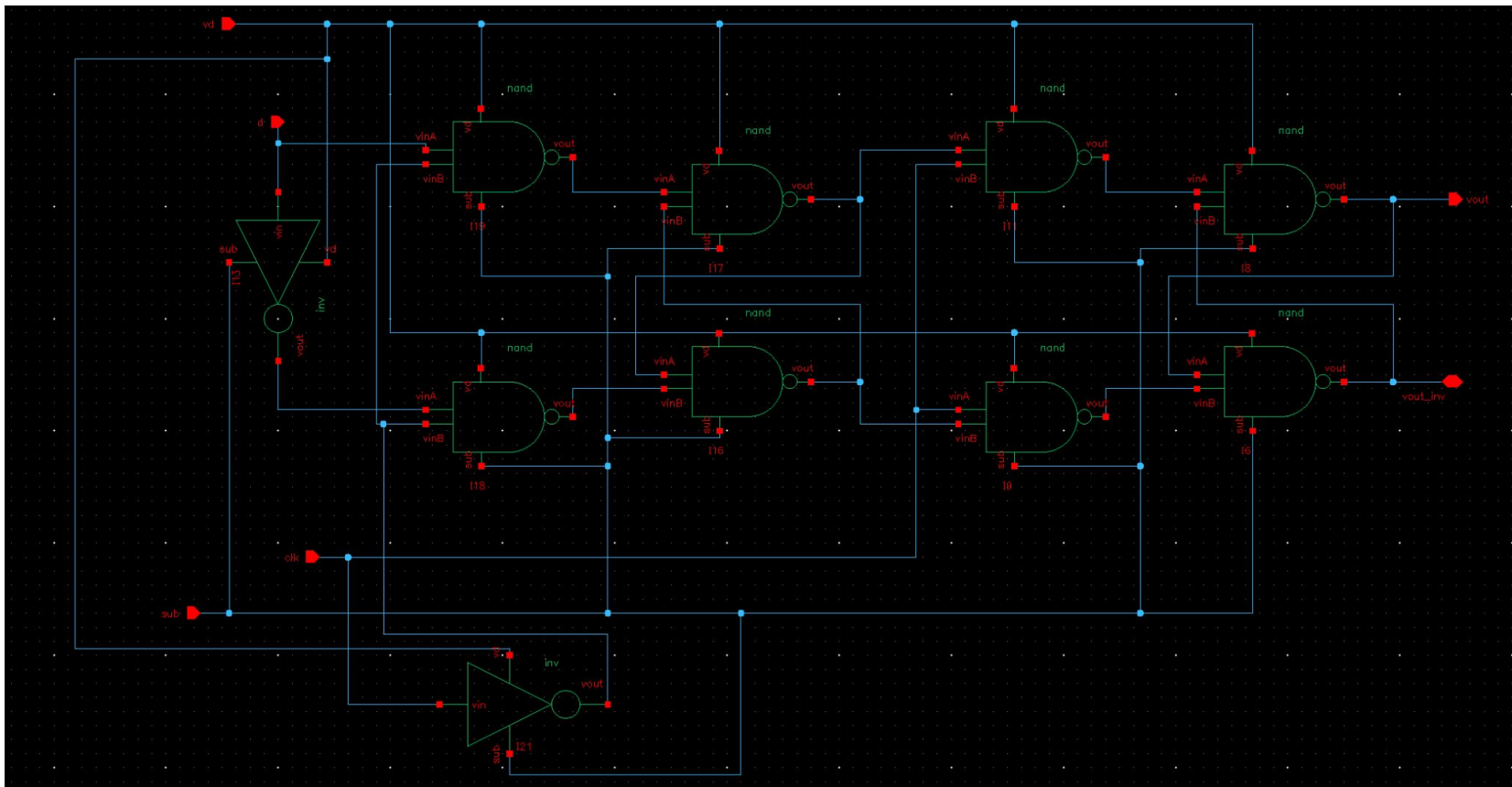
Run Directory: /home/eceguest/digital_course/LVS

```

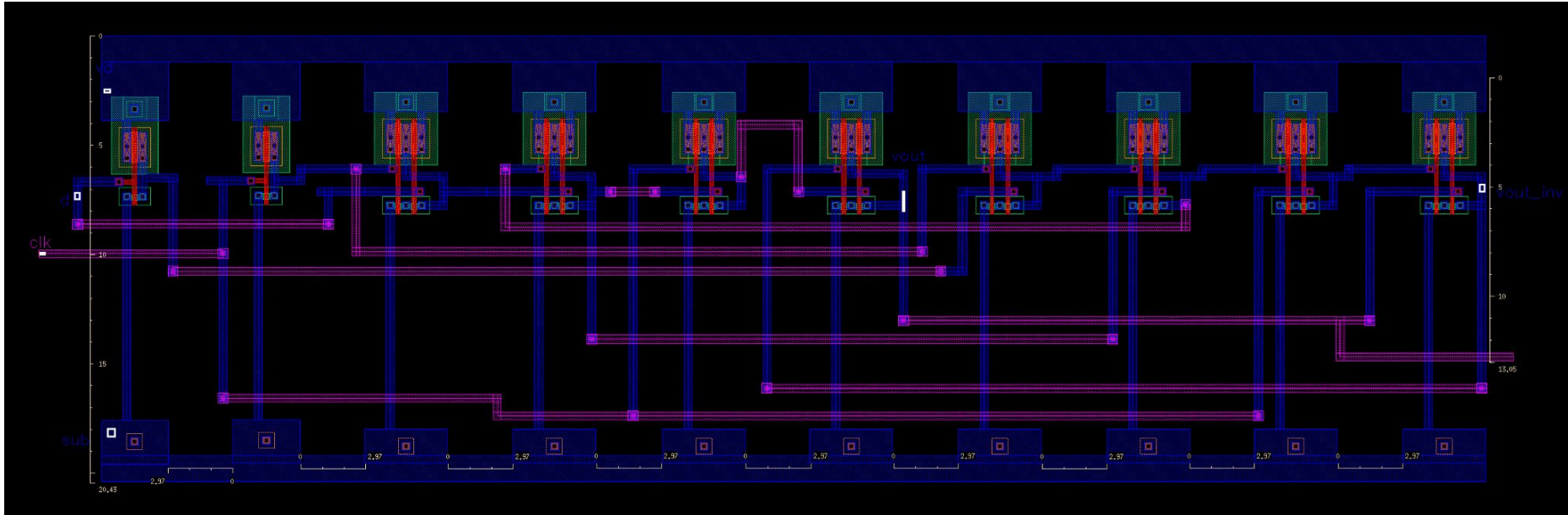
Xor gate LVS



D-FlipFlop



D-FlipFlop Schematic



D-FlipFlop Layout

```
DRC started at Wed Dec 6 21:10:22 2023
```

```
Validating hierarchy instantiation for:
```

```
library: Stanford
```

```
cell: D_flipflop
```

```
view: layout
```

```
Rules come from library NCSU_TechLib_tsmc02d.
```

```
Rules path is divaDRC.rul.
```

```
Inclusion limit is set to 1000.
```

```
Running layout DRC analysis
```

```
Flat mode
```

```
Full checking.
```

```
DRC started.....Wed Dec 6 21:10:22 2023
```

```
completed ....Wed Dec 6 21:10:22 2023
```

```
CPU TIME = 00:00:00 TOTAL TIME = 00:00:00
```

```
***** Summary of rule violations for cell "D_flipflop layout" *****
```

```
Total errors found: 0
```

D-FlipFlop DRC

Extraction started at Wed Dec 6 21:11:06 2023

Validating hierarchy instantiation for:

library: Stanford

cell: D_flipflop

view: layout

Rules come from library NCSU_TechLib_tsmc02d.

Rules path is divaEXT.rul.

Inclusion limit is set to 1000.

Running layout Extraction analysis

flat mode

Full checking.

For layer nChannelTran :

18 shapes encountered.

18 nmos4 ivpcell NCSU_Analog_Parts devices well formed.

For layer pChannelTran :

18 shapes encountered.

18 pmos4 ivpcell NCSU_Analog_Parts devices well formed.

0 cap ivpcell NCSU_Analog_Parts parasitics created.

0 cap ivpcell NCSU_Analog_Parts parasitics created.

0 cap ivpcell NCSU_Analog_Parts parasitics created.

0 cap ivpcell NCSU_Analog_Parts parasitics created.

0 cap ivpcell NCSU_Analog_Parts parasitics created.

0 cap ivpcell NCSU_Analog_Parts parasitics created.

Extraction started.....Wed Dec 6 21:11:06 2023

completed ...Wed Dec 6 21:11:07 2023

CPU TIME = 00:00:00 TOTAL TIME = 00:00:01

***** Summary of rule violations for cell "D_flipflop layout" *****

Total errors found: 0

saving rep Stanford/D_flipflop/extracted

Getting layout proper bagGetting layout proper bag

Error eval: unbound variable - ivLVSContentsDifferForm

LVS job is now started...

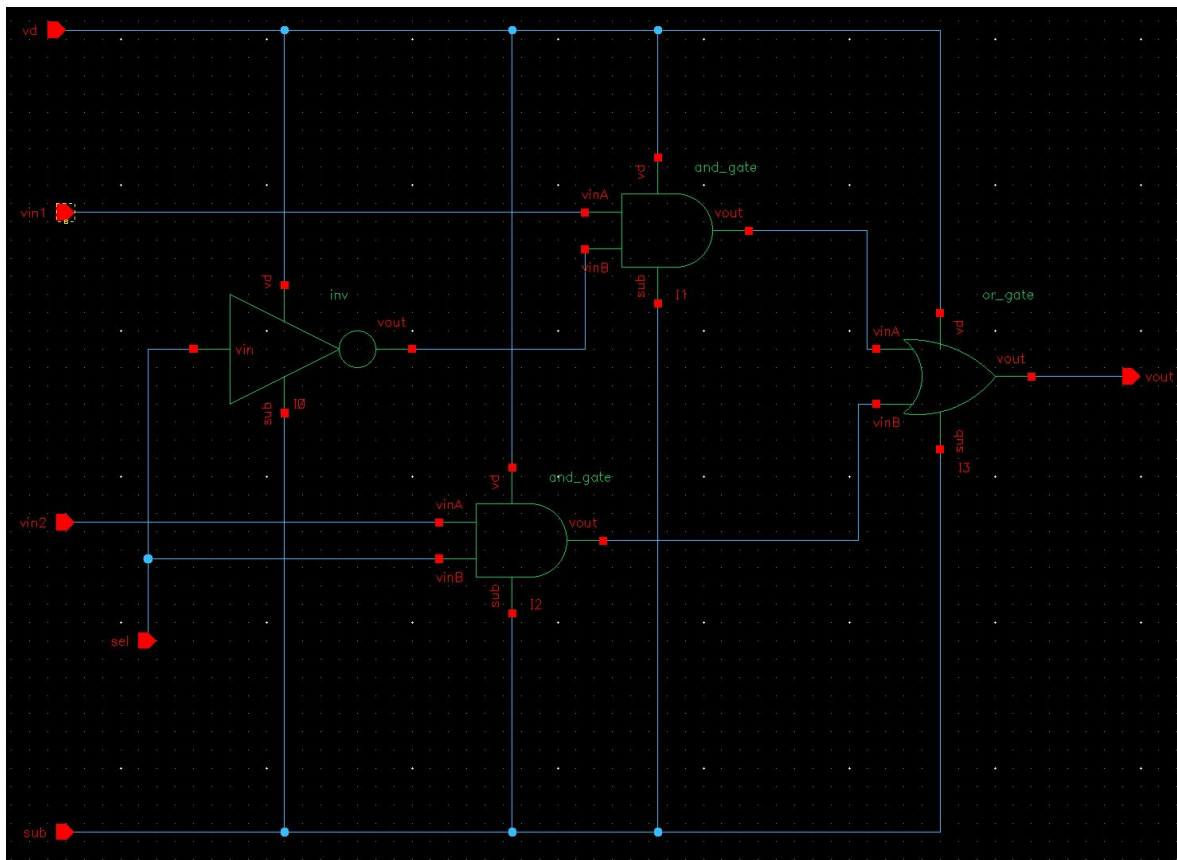
The LVS job has completed. The net-lists match.

Run Directory: /home/eceguest/digital_course/LVS

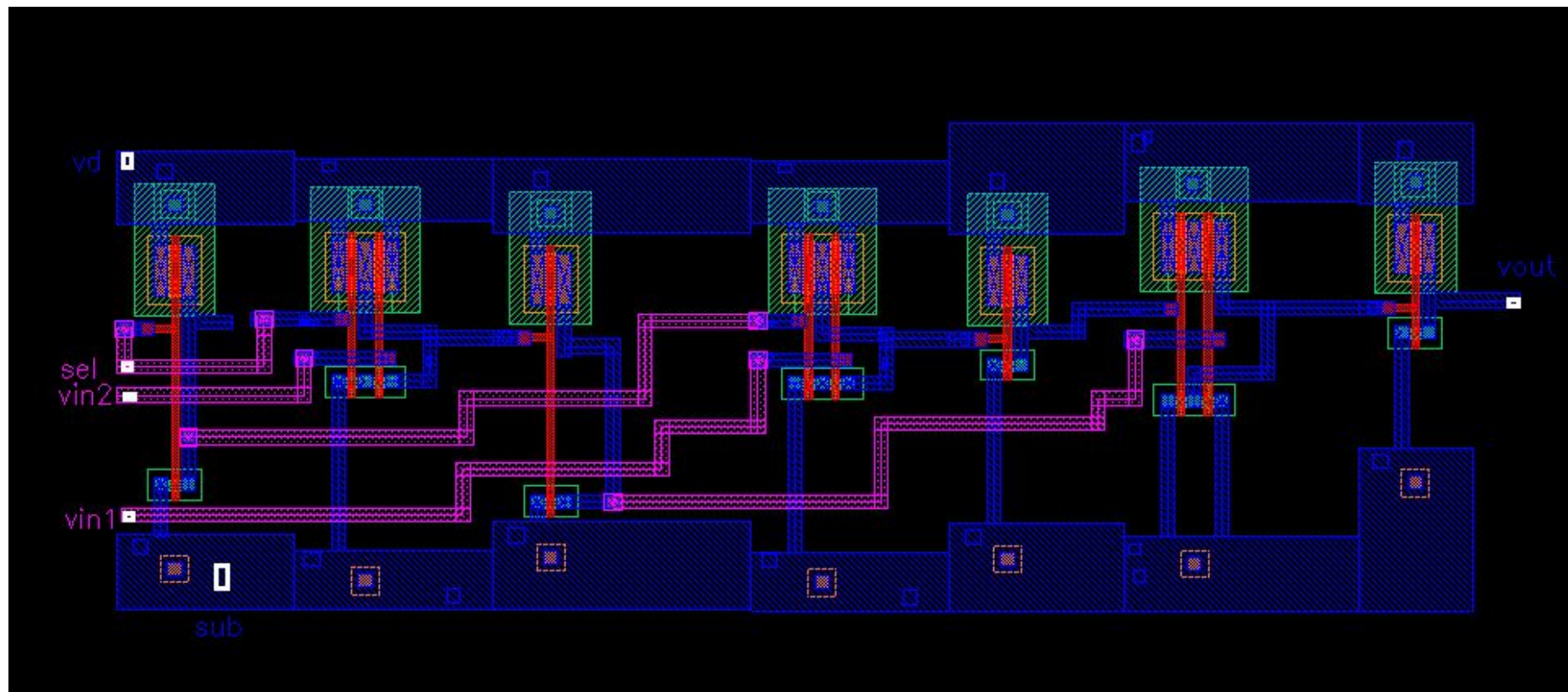
D-FlipFlop LVS



2-to-1 mux



2-to-1 mux Schematic



2-to-1 mux Layout

DRC started at Wed Dec 6 21:26:15 2023

Validating hierarchy instantiation for:

library: Stanford

cell: 2-to-1mux

view: layout

Rules come from library NCSU_TechLib_tsmc02d.

Rules path is divaDRC.rul.

Inclusion limit is set to 1000.

Running layout DRC analysis

Flat mode

Full checking.

DRC started.....Wed Dec 6 21:26:15 2023

completedWed Dec 6 21:26:15 2023

CPU TIME = 00:00:00 TOTAL TIME = 00:00:00

***** Summary of rule violations for cell "2-to-1mux layout" *****

Total errors found: 0

2-to-1 mux DRC

```

Extraction started at Wed Dec 6 21:26:50 2023

Validating hierarchy instantiation for:
library: Stanford
cell: 2-to-1mux
view: layout
Rules come from library NCSU_TechLib_tsmc02d.
Rules path is divaEXT.rul.
Inclusion limit is set to 1000.
Running layout Extraction analysis
flat mode
Full checking.
For layer nChannelTran :
  10 shapes encountered.
  10 nmos4 ivpcell NCSU_Analog_Parts devices well formed.
For layer pChannelTran :
  10 shapes encountered.
  10 pmos4 ivpcell NCSU_Analog_Parts devices well formed.
0 cap ivpcell NCSU_Analog_Parts parasitics created.
0 cap ivpcell NCSU_Analog_Parts parasitics created.
0 cap ivpcell NCSU_Analog_Parts parasitics created.
0 cap ivpcell NCSU_Analog_Parts parasitics created.
0 cap ivpcell NCSU_Analog_Parts parasitics created.
0 cap ivpcell NCSU_Analog_Parts parasitics created.
Extraction started.....Wed Dec 6 21:26:50 2023
  completed ....Wed Dec 6 21:26:50 2023
  CPU TIME = 00:00:00 TOTAL TIME = 00:00:00
***** Summary of rule violations for cell "2-to-1mux layout" *****
Total errors found: 0

saving rep Stanford/2-to-1mux/extracted
Getting layout propert bagGetting layout propert bag
*Error* eval: unbound variable - ivLVSContentsDifferForm
LVS job is now started...
The LVS job has completed. The net-lists match.

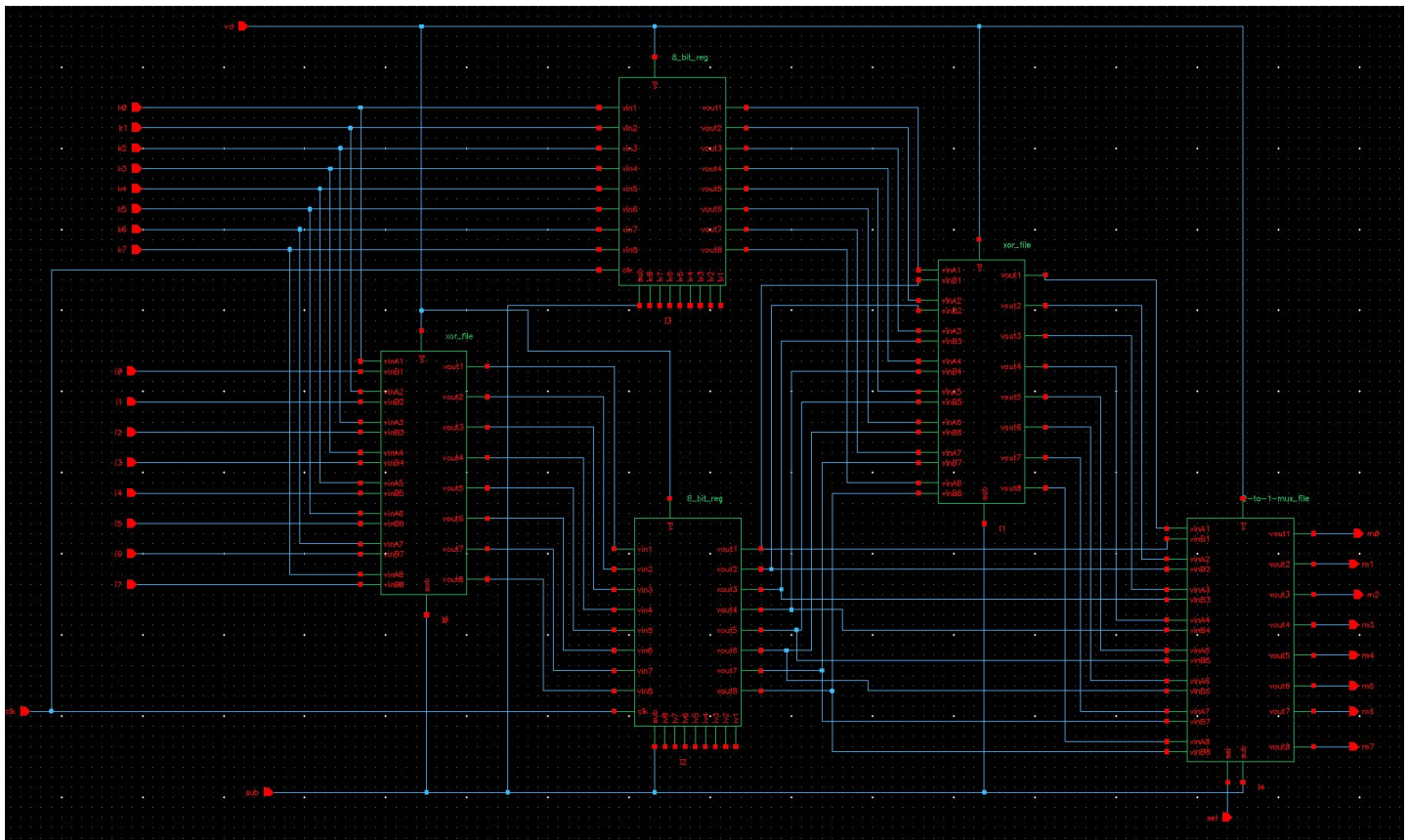
Run Directory: /home/eceguest/digital_course/LVS

```

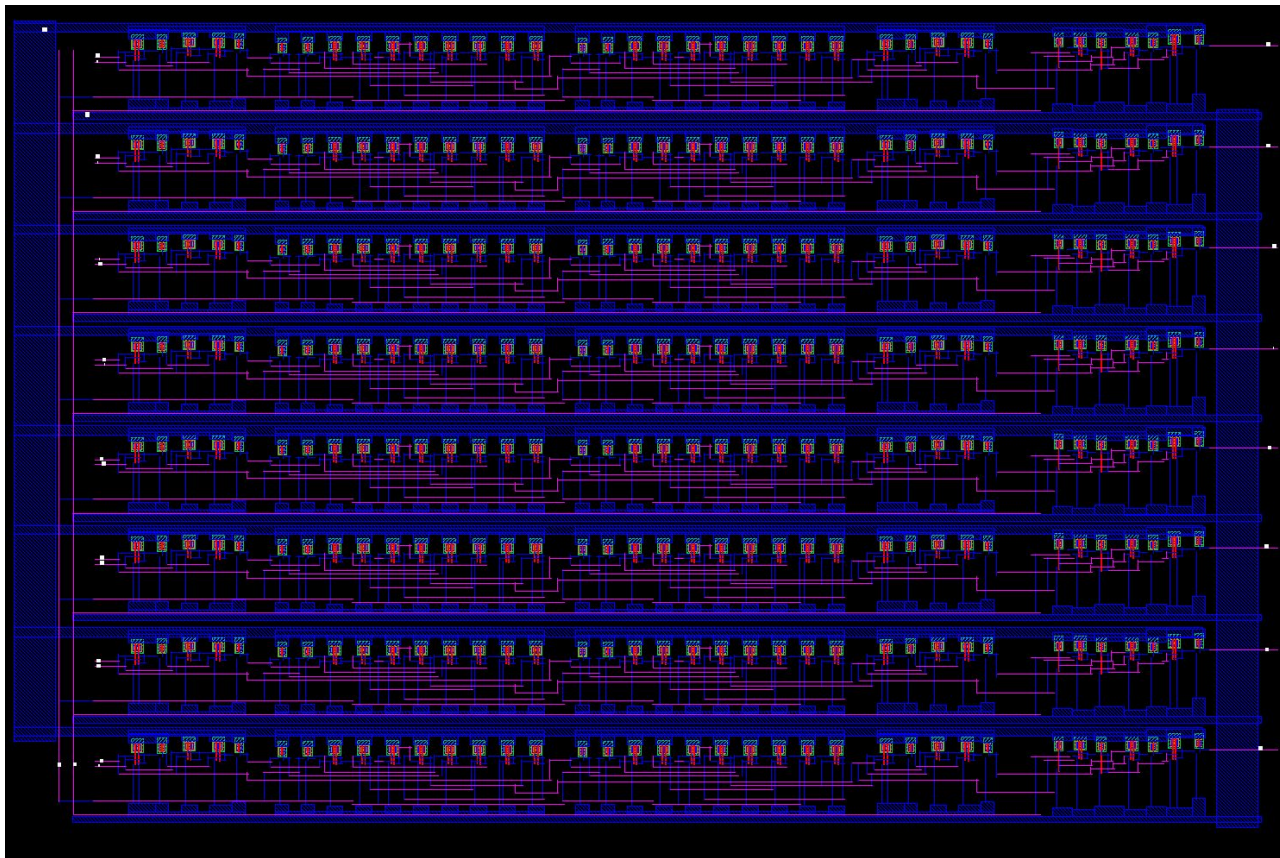
2-to-1 mux LVS



Encryptor



Encryptor Schematic



Encryptor Layout

DRC started at Wed Dec 6 21:32:13 2023

Validating hierarchy instantiation for:

library: Stanford

cell: xor_encryptor_decryptor

view: layout

Rules come from library NCSU_TechLib_tsmc02d.

Rules path is divaDRC.rul.

Inclusion limit is set to 1000.

Running layout DRC analysis

Flat mode

Full checking.

DRC started.....Wed Dec 6 21:32:13 2023

completedWed Dec 6 21:32:14 2023

CPU TIME = 00:00:00 TOTAL TIME = 00:00:01

***** Summary of rule violations for cell "xor_encryptor_decryptor layout" *****

Total errors found: 0

Encryptor DRC

Extraction started at Wed Dec 6 21:33:19 2023

Validating hierarchy instantiation for:

library: Stanford

cell: xor_encryptor_decryptor

view: layout

Rules come from library NCSU_TechLib_tsmc02d.

Rules path is divaEXT.rul.

Inclusion limit is set to 1000.

Running layout Extraction analysis

flat mode

Full checking.

For layer nChannelTran :

496 shapes encountered.

496 nmos4 ivpcell NCSU_Analog_Parts devices well formed.

For layer pChannelTran :

496 shapes encountered.

496 pmos4 ivpcell NCSU_Analog_Parts devices well formed.

0 cap ivpcell NCSU_Analog_Parts parasitics created.

0 cap ivpcell NCSU_Analog_Parts parasitics created.

0 cap ivpcell NCSU_Analog_Parts parasitics created.

0 cap ivpcell NCSU_Analog_Parts parasitics created.

0 cap ivpcell NCSU_Analog_Parts parasitics created.

0 cap ivpcell NCSU_Analog_Parts parasitics created.

Extraction started.....Wed Dec 6 21:33:19 2023

completed ...Wed Dec 6 21:33:20 2023

CPU TIME = 00:00:00 TOTAL TIME = 00:00:01

***** Summary of rule violations for cell "xor_encryptor_decryptor layout" *****

Total errors found: 0

saving rep Stanford/xor_encryptor_decryptor/extracted

Getting layout proper bagGetting layout proper bag

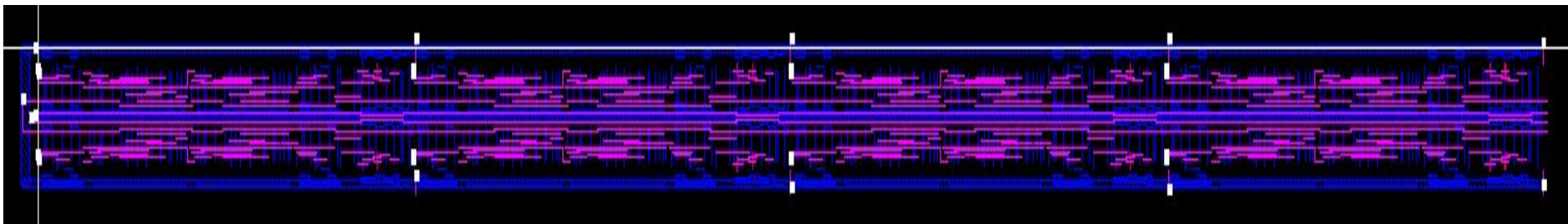
LVS job is now started...

The LVS job has completed. The net-lists match.

Encryptor LVS



Issues / Improvements



DRC started at Wed Dec 6 22:34:10 2023

Validating hierarchy instantiation for:

library: Stanford

cell: xor_encryptor_decryptor_2

view: layout

Rules come from library NCSU_TechLib_tsmc02d.

Rules path is divaDRC.rul.

Inclusion limit is set to 1000.

Running layout DRC analysis

Flat mode

Full checking.

DRC started.....Wed Dec 6 22:34:10 2023

completedWed Dec 6 22:34:11 2023

CPU TIME = 00:00:00 TOTAL TIME = 00:00:01

***** Summary of rule violations for cell "xor_encryptor_decryptor_2 layout" *****

Total errors found: 0

Encryptor (More compact) Layout / DRC

Extraction started at Wed Dec 6 22:34:49 2023

Validating hierarchy instantiation for:

library: Stanford
cell: xor_encryptor_decryptor_2
view: layout
Rules come from library NCSU_TechLib_tsmc02d.
Rules path is divaEXT.rul.

Inclusion limit is set to 1000.
Running layout Extraction analysis
flat mode
Full checking.

For layer nChannelTran :
496 shapes encountered.
496 nmos4 ivpcell NCSU_Analog_Parts devices well formed.
For layer pChannelTran :
496 shapes encountered.
496 pmos4 ivpcell NCSU_Analog_Parts devices well formed.

0 cap ivpcell NCSU_Analog_Parts parasitics created.
0 cap ivpcell NCSU_Analog_Parts parasitics created.
0 cap ivpcell NCSU_Analog_Parts parasitics created.
0 cap ivpcell NCSU_Analog_Parts parasitics created.
0 cap ivpcell NCSU_Analog_Parts parasitics created.
0 cap ivpcell NCSU_Analog_Parts parasitics created.

Extraction started.....Wed Dec 6 22:34:49 2023
completed ...Wed Dec 6 22:34:50 2023

CPU TIME = 00:00:00 TOTAL TIME = 00:00:01
***** Summary of rule violations for cell "xor_encryptor_decryptor_2 layout" *****
Total errors found: 0

saving rep Stanford/xor_encryptor_decryptor_2/extracted

Getting layout proper bagGetting layout proper bag

Error eval: unbound variable - ivLVSContentsDifferForm

LVS job is now started...

WARNING The LVS job has failed to run to completion. Examine si.out and si.log in the run directory for more information.

Run Directory: /home/eceguest/digital_course/LVS

LVS job is now started...

WARNING The LVS job has failed to run to completion. Examine si.out and si.log in the run directory for more information.

Run Directory: /home/eceguest/digital_course/LVS

LVS job is now started...

The LVS job has completed. The net-lists failed to match.

Run Directory: /home/eceguest/digital_course/LVS

WARNING (LE-104054): Stretch expansion is limited to 5000 wiring objects. Connectivity may not be maintained.

The net-lists failed to match.

	layout	schematic
	instances	
un-matched	0	0
rewired	4	0
size errors	0	0
pruned	0	0
active	992	992
total	992	992

	nets	
un-matched	0	0
merged	0	0
pruned	0	0
active	516	516
total	516	516

	terminals	
un-matched	0	0
matched but		
different type	0	0
total	28	28

End comparison: Dec 6 22:36:26 2023

Comparison program completed successfully.

Encryptor (More compact) LVS - Fail / LVS log



Improvements to be made

- Reduce wasted space in the individual components layout
- More Compact Overall layout, without current issues
- Reduction of gates in the path and better design
 - Flip Flop in particular



Notes

Move the nwell closer to the transistors

- specifically the orange boxes at the base

Basically move the nmos closer to the nwell,

But should still be able to route